DataOpus
discover, learn, improve

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RFI Response

Secondary Title

The DataOpus Solution

CMS, state governments, and commercial organizations producing healthcare solutions should incorporate DataOpus into their engineering processes to quickly and confidently detect vulnerabilities, weaknesses, and data integrity issues and provide courses of action to remedy them.

Our knowledge of CMS and working with state health-based agencies were a leading reason why DataOpus was created. In less than 2-minutes DataOpus;

1. Produces actionable findings preventing the database portion of a healthcare system from being HIPAA compliant (vulnerabilities, weaknesses, flaws, data integrity issues, etc.)
2. Creates a data dictionary to help teams with maintenance, interoperability, reports, AI, etc.
3. Creates a code inventory to help with maintenance and interoperability
4. Measures & Metrics
5. List of actionable risks, opportunities for improvement, and waste to include best practices, learning opportunities, and how to address each
6. Reduces 99% of the time and cost to find issues so organizations can dedicate that time to resolving them

Responses to Relevant Questions

1. Patient Needs

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

a. What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)? What criteria should be used if there is a review process? What technology solutions,

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policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)?

Minimally, CMS should require self-attestation of items listed in the database warranty. Ideally, CMS would require use of a 3rd party tool to determine health of their data platforms.

b. What changes would enable timely access to high quality CMS and provider generated data on patients?

Require the use of data dictionaries and descriptions of properties sent to and received from web services to increase interoperability.

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

1. Require full transparency of the measures provided & metrics to include data provenance and algorithms
2. Request use cases
3. Request
4. Require highly interoperable formats for interoperability and future proofing
5. Require data formats that can handle internationalization / time zone differences

2. Data Access and Integration

c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?

1. Effective data dictionaries
2. Date handling for all time zones
3. Alignment to healthcare related policies, standards, regulations
4. Future-proofed data architectures

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5. Systemic data integrity checks

PC-9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?

a. What difficulties are there in accessing or utilizing these data sources today?

Understanding the primary purpose of data elements, potential values, data dictionaries, APIs, etc.

b. What suggestions do you have to improve the Blue Button 2.0 API experience?

- Require extensive data dictionaries
- Require fully descriptive APIs to include fully describing the parameters and results

PC-11. How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world?

b. What changes would you suggest?

1. Highly detailed data dictionaries
2. Highly detailed extendable APIs

3. Information Blocking and Digital Identity

PC-14. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 credentialing service providers (CSP)):

a. What are the challenges today in getting patients/caregivers to sign up and use digital identity credentials?

1. Commitment to making MFA simple yet efficient
2. Commitment to keeping data on US-based data servers
3. Commitment to anonymity beyond initial capture

b. What could be the benefits to patients/caregivers if digital identity credentials were more widely used?

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1. Capture more information more quickly
2. Opportunity for a more 360-data approach

c. What are the potential downsides?

1. Further loss of trust like we seen with the Obamacare disaster
2. Lack of participation

d. How would encouraging the use of CSPs improve access to health information?

1. Provides opportunity for quicker and more frequent communication
2. Provides opportunity to address data / privacy information

e. What role should CMS/payers, providers, and app developers have in driving adoption?

1. Their User Experience (UC) approach would either hinder or help adoption.
2. Data usage / terms of use policy easily understood by an 8th grader (seriously)

f. How can CMS encourage patients to get digital identity credentials?

1. Clearly explain the value to them personally by using non-legal jargon

C. Providers

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve providers. While we certainly want providers to answer questions in this section (and in other sections) from the provider point of view, we also invite all stakeholders to provide their viewpoints on the provider workflows as appropriate.

1. Digital Health Apps

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?

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a. What are the current obstacles?

1. Internet access
2. Efficient devices

b. What information should providers share with patients when using digital products in the provision of their care?

1. Duh – All

c. What responsibilities do providers have when recommending use of a digital product by a patient?

1. Data privacy

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

1. The big 4
2. Dedication to User Experience
3. Low-end software engineers
4. IT acquisition for small businesses to be a disruptor

PR-3. How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)? Please address all of the following:

a. Current challenges in accessing different data formats.

1. Too much freeform text: need to find more ways to standardize data via pattern recognition
2. Unsure of completeness and quality of data

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b. Impact on patient care quality.

c. Technical barriers to full data accessibility.

1. Standardized encryption with data at rest and in-flight

d. Cost or privacy implications of making all data formats interoperable.

1. Without know what data elements in formats should be encrypted, in cases when not fully encrypted data leakage is possible

e. Priority level compared to other interoperability needs.

1. After IAM, high priority as we want secure higher quality data

PR-4. What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?

1. All organizations with data should have continual 3rd party audits on its security, privacy, and quality

2. Data Exchange

PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?

1. Require system forensic analysis of all data platforms that would provide results on non-functional requirements like interoperability, performance, usability, etc.
2. Utilize the NIST framework for risk management for consistency across all CMS related systems
3. Require data dictionaries

PR-8. What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?

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1. Host workshops on this subject

c. Are there requirements CMS should consider for data registries to support digital quality measurement in a more efficient manner? Are there requirements CMS should consider for data registries that would support access to real-time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

1. Systemic forensic analysis by 3rd party

E. Technology Vendors, Data Providers, and Networks

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve technology vendors, data providers, and networks. While we certainly want technology vendors, data providers, and networks to answer questions in this section (and in other sections) from their point of view, we also invite all stakeholders to provide their viewpoints on the technology vendor, data provider, and network use cases as appropriate.

1. Ecosystem

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

1. Biggest issue is the IT acquisition process that keeps innovators out of the market

4. Data Exchange

TD-12. Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

1. Must know the provenance of data
2. 3rd party systemic evaluations for Data-readiness, AI-readiness
3. Alignment to CMS data standards

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

1. Greater trust in data could lead to more application of AI/ML

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a. What are the primary obstacles to this?

1. Data volume, provenance, and quality